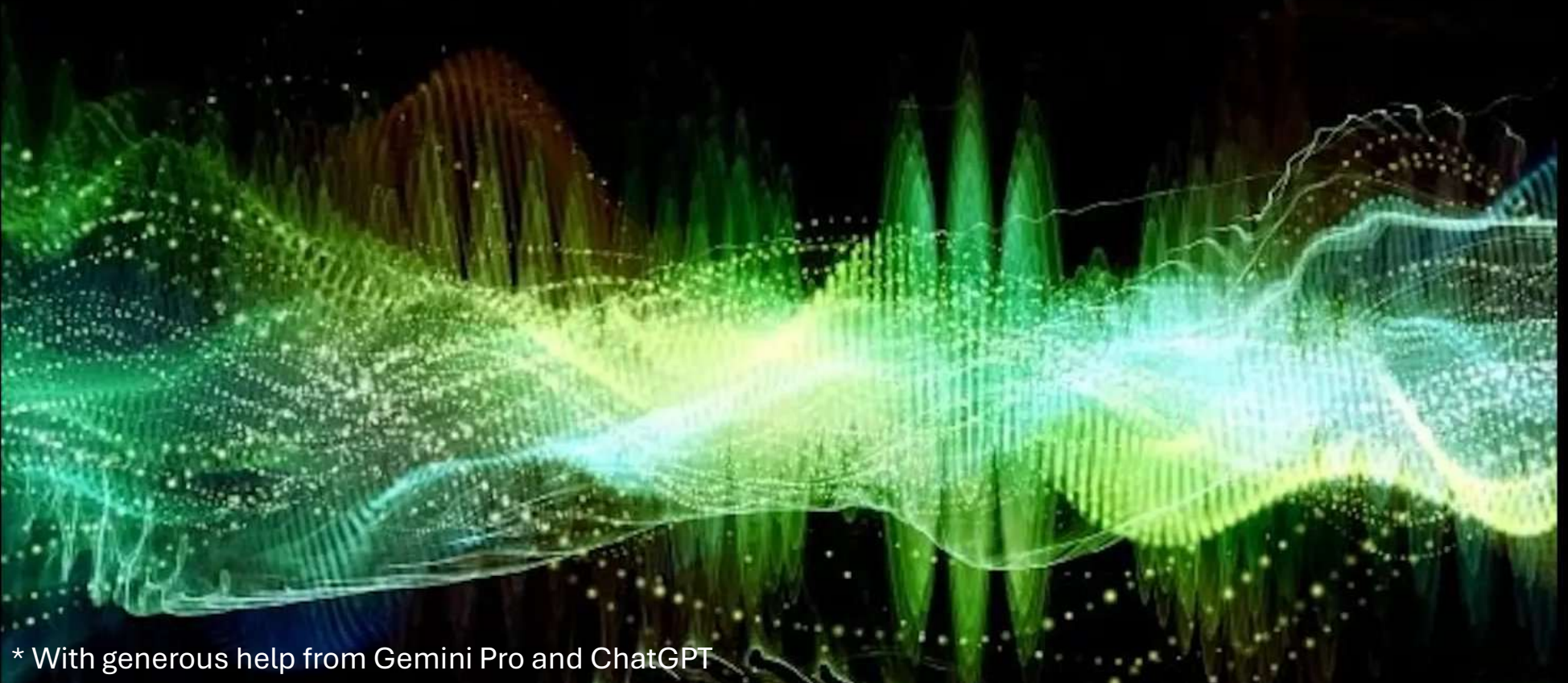


ECEN 4/5632 Theory and Application of Digital Filters

Lecture 5, 9-08-26

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* With generous help from Gemini Pro and ChatGPT

The z-Transform and the Region of Convergence

Beyond the DTFT: convergence becomes geometry.

**DTFT = z-transform
on the unit circle**

75 minutes • rigorous DSP foundations • audio examples • Python sandbox • ML bridge

Learning outcomes

00–15 MIN

By the end, students should reason from both algebra and ROC geometry.

- Explain why the DTFT can fail while a z-transform exists elsewhere in the z-plane.
- Compute bilateral z-transforms of common right- and left-sided sequences.
- Use the ROC to distinguish sequences with the same rational expression.
- Perform inverse z-transforms by partial fraction expansion.
- Connect unit-circle inclusion to the existence of the DTFT.

Beyond the DTFT: sometimes the unit circle is the problem

00-15 MIN

The DTFT is only one radial slice of a larger transform.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] \cdot z^{-n} \quad \text{let } z^n = r^n e^{j\omega n}$$

Example

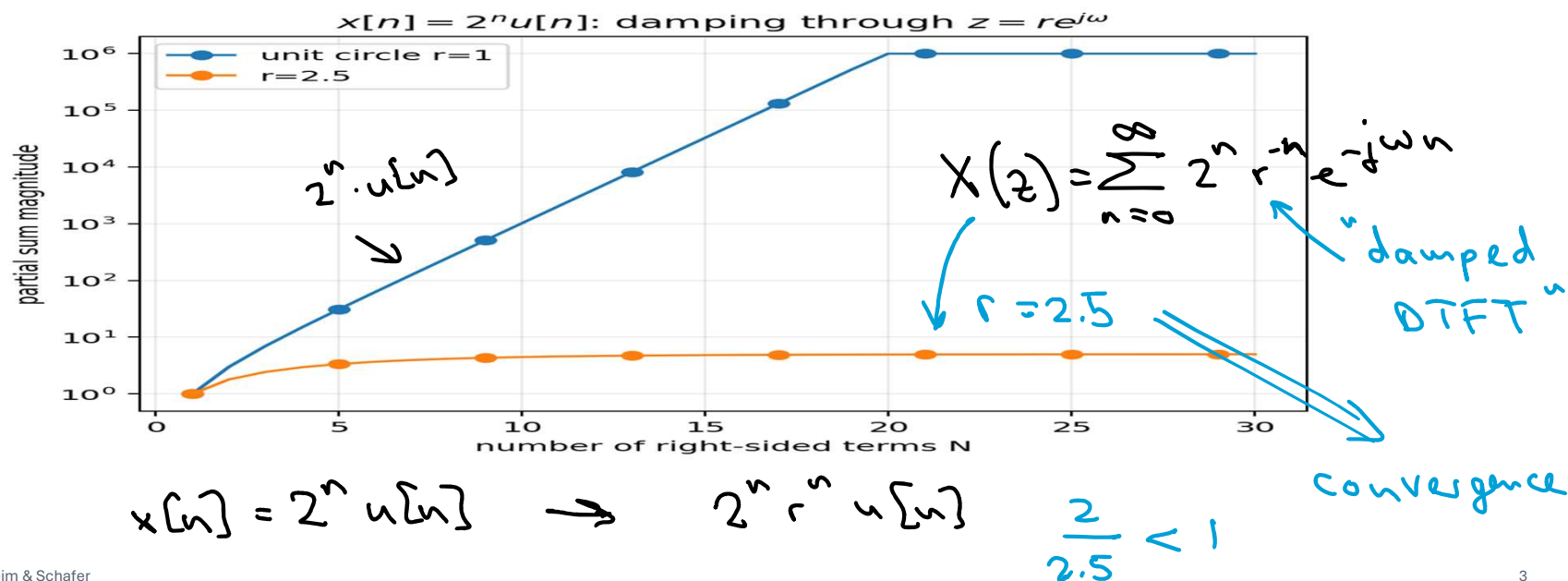
$x[n] = 2^n u[n]$ grows too quickly.

On the unit circle

$\sum 2^n e^{-j\omega n}$ does not converge in the ordinary sense.

Move radially outward

$z^{-n} = r^{-n} e^{-j\omega n}$ adds damping. For $r > 2$, $(2/r)^n$ decays.

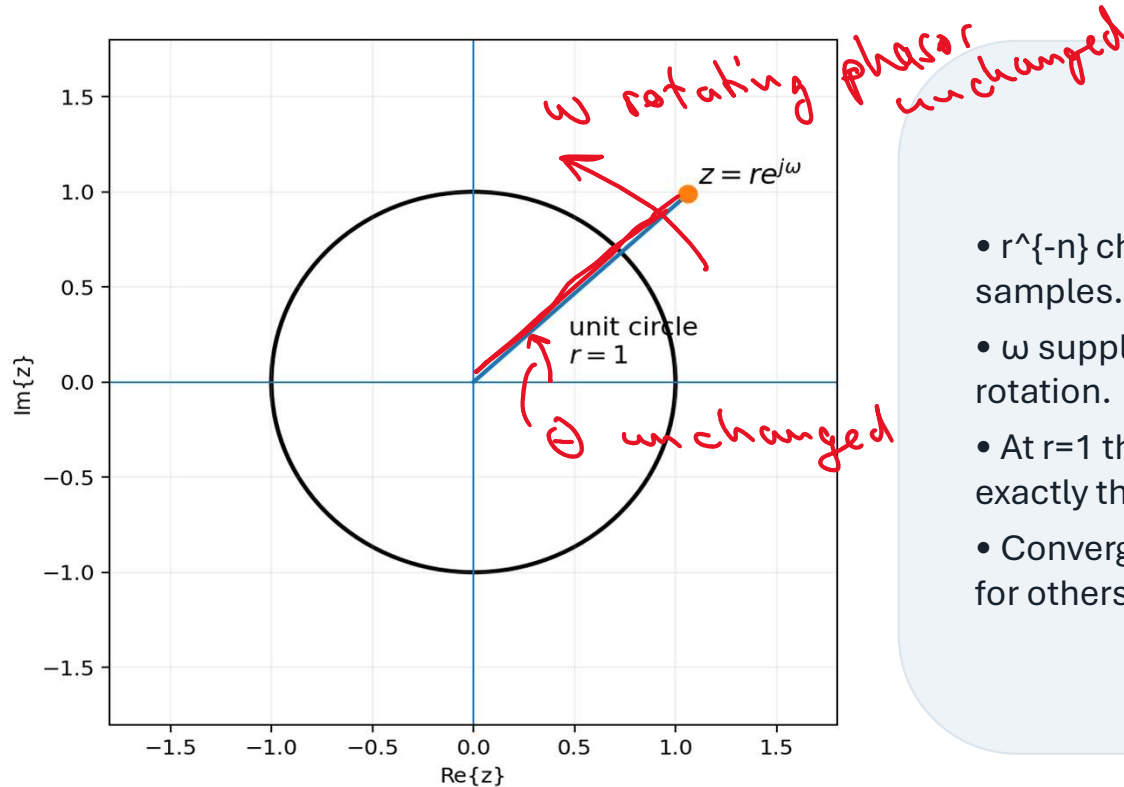


The damping-factor interpretation

Write z in polar form: radius controls convergence; angle probes frequency.

$$X(z) = \sum_n x[n] z^{-n}$$

00-15 MIN



$$z = r e^{j\omega}$$

- r^{-n} changes the weighting of positive-time samples.
- ω supplies the familiar complex sinusoidal rotation.
- At $r=1$ there is no radial damping: this is exactly the DTFT.
- Convergence can hold for some radii and fail for others.

if $r = 1$: no radial damping
 $r > 1 \Rightarrow$ radial damping "reiging in $x[n]$ "

Definition of the bilateral z-transform

15-35 MIN

A transform value exists only where the infinite sum converges.

$$X(z) = \sum_{n=-\infty}^{\infty} x[n]z^{-n}$$

Complex variable

z can be anywhere in the complex plane—not just on the unit circle.

Transform + ROC

A rational formula without its ROC is incomplete information.

Bilateral

Both positive and negative n may contribute; this is why annular ROCs appear.

$$X(z) = \frac{B(z)}{A(z)}$$

← numerator
← denominator

rational function

in addition ROC is needed to make $X(z)$ unique

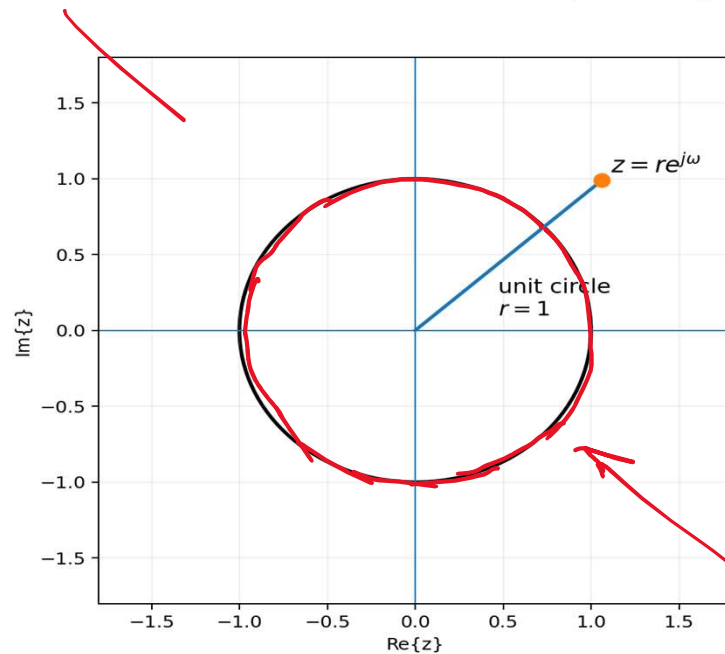
The DTFT lives on the unit circle

15-35 MIN

The z-transform generalizes frequency analysis radially.

$\sqrt{z}$ plane

$$X(e^{j\omega}) = X(z) \text{ at } z = e^{j\omega}$$



- If the ROC includes $|z|=1$, the DTFT exists in the ordinary sense.
- Move around the unit circle: $z=e^{j\omega}$.
- A pole on the unit circle prevents a finite frequency response at that angle.
- Later: stability of an LTI system is tied to the unit circle lying in the ROC.

DTFT is evaluated on unit circle
in the z-plane

Standard pair: right-sided exponential

15–35 MIN

Derive it as a geometric series—do not memorize the ROC separately.

$$a^n u[n] \leftrightarrow \frac{1}{1 - az^{-1}}, \quad \underline{|z| > |a|}$$

Derivation

$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \sum_{n=0}^{\infty} (az^{-1})^n$$

Geometric series converges when $|az^{-1}| < 1$
 $\Rightarrow |z| > |a|$

Interpretation

- Pole at $z=a$.
- ROC extends outward from the pole.
- Right-sided sequences naturally produce exterior ROCs.
- If $|a| < 1$, the unit circle lies in the ROC and the DTFT exists.

$$x[n] = a^n u[n] \Rightarrow X(z) = \sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1 - az^{-1}}$$

↑
if $|az^{-1}| < 1 \Rightarrow |z| > |a|$

$$\underline{X(z)} = \frac{1}{1 - az^{-1}} \quad |z| > |a|$$

$$x[n] = a^n \cdot u[n]$$

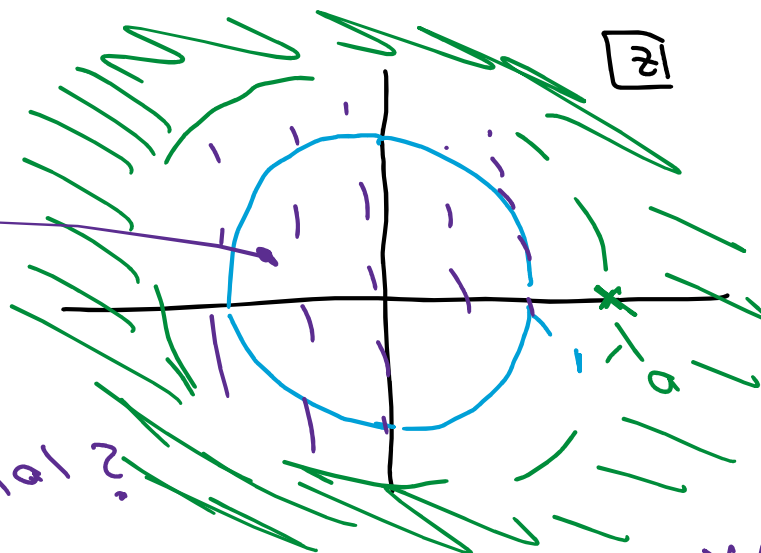
long division

$$1 + az^{-1} + a^2z^{-2} + \dots$$

$$\begin{array}{r} 1 - az^{-1} \overline{) 1} \\ \underline{1 - az^{-1}} \\ az^{-1} \\ \underline{az^{-1} - a^2z^{-2}} \\ a^2z^{-2} \end{array}$$

what happens for $|z| < |a|$?

another ROC for same $X(z)$



$$X(z) = \sum_n x[n] z^{-n} \quad \leftarrow \text{by convention} \quad \text{why not } \sum_n x[n] z^n$$

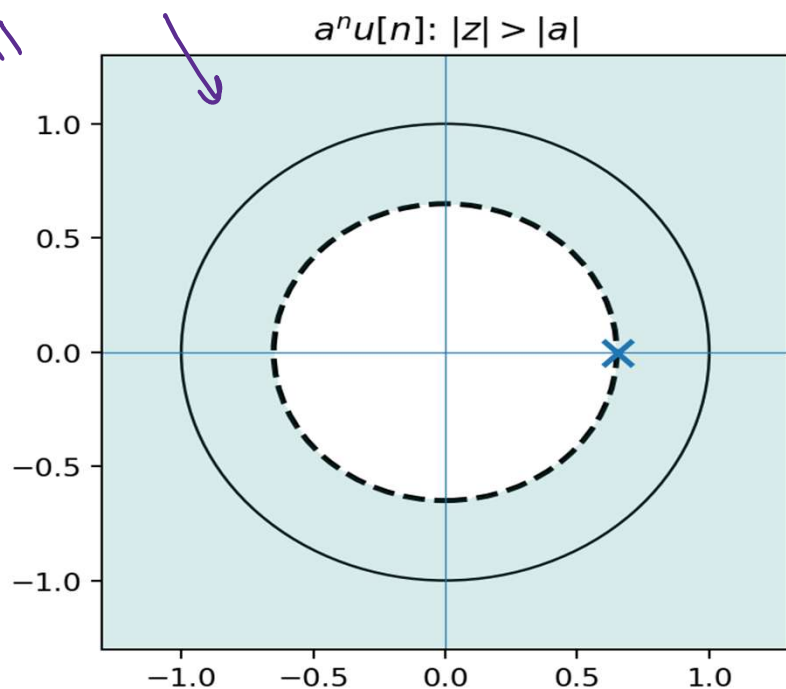
$n - n_0$ represents a delay
whereas $n + n_0$ represents advance

Same rational expression, different sequence

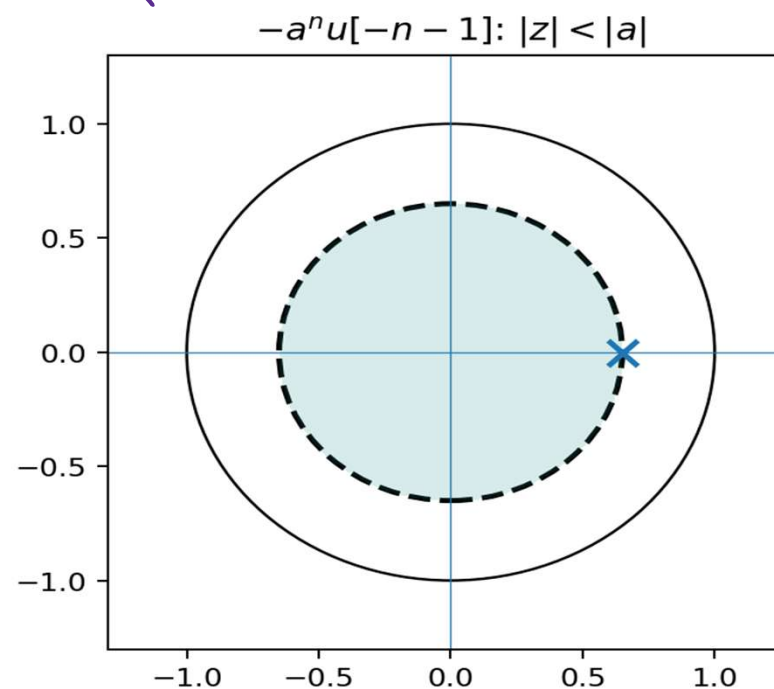
35-55 MIN

The ROC is what tells you which time support is intended.

*ROC
 $|z| > |a|$*



*other ROC
 $|z| < |a|$*



Key fact: $1/(1-az^{-1})$ alone does NOT uniquely determine $x[n]$. The ROC is part of the transform.

First ROC $|z| > |a|$ $x[n] = a^n \cdot u[n]$ for $n \in [0, \infty]$

Another portion of n is $n \in [-\infty, -1]$

$$\sum_{n=-\infty}^{-1} a^n z^{-n} = \sum_{m=\infty}^1 a^{-m} z^m = \sum_{m=1}^{\infty} a^{-m} z^m = \frac{1}{1 - a^{-1}z} - 1 = \frac{1 - 1 + a^{-1}z}{1 - a^{-1}z}$$

$|a^{-1}z| < 1 \Rightarrow \text{ROC } |z| < |a|$

$$\overset{\substack{-m \leftarrow n \\ m \leftarrow -n}}{=} \frac{a^{-1}z}{1 - a^{-1}z} \overset{\substack{\uparrow \\ \text{times } a \cdot z^{-1}}}{=} \frac{1}{az^{-1} - 1} = \frac{-1}{1 - az^{-1}}$$

$$x_1[n] = a^n \cdot u[n] \Leftrightarrow X_1(z) = \frac{1}{1 - az^{-1}} \quad |z| > |a| \text{ ROC}_1$$

same z-transform

$$x_2[n] = -a^n u[-n-1] \Leftrightarrow X_2(z) = \frac{+1}{1 - az^{-1}} \quad |z| < |a| \text{ ROC}_2$$

Long division starting with highest powers of z

$$1 + az^{-1} + a^2z^{-2} + \dots$$

$$\begin{array}{r} 1 - az^{-1} \overline{) 1} \\ \underline{1 - az^{-1}} \\ \vdots \end{array}$$

← right-sided sequence expansion
 causal system if $x[n] = h[n]$
 Laurent series
 ROC $|z| > |a|$

Long division starting with lowest powers of z first

$$-a^{-1}z - a^{-2}z^2 - a^{-3}z^3 - \dots \quad x_2[n]$$

$$\begin{array}{r} -a^{-1}z + 1 \overline{) 1} \\ \underline{1 - a^{-1}z} \\ a^{-1}z \\ \underline{a^{-1}z - a^{-2}z^2} \\ a^{-2}z^2 \end{array}$$

← left-sided sequence expansion
 anticausal system if $x[n] = h[n]$
 Taylor series
 ROC $|z| < |a|$

Left-sided exponential: why the minus sign appears

35-55 MIN

Expand the same rational function in powers appropriate to $|z| < |a|$.

For $|z| < |a|$ *ROC
anti-causal
sequence*

$$1/(1-az^{-1}) = z/(z-a)$$

Rewrite so the geometric ratio has magnitude < 1 :

$$\begin{aligned} &= -(z/a)/(1-z/a) \\ &= -\sum_{k=1}^{\infty} (z/a)^k \end{aligned}$$

Inverse sequence

$$-a^n u[-n-1]$$

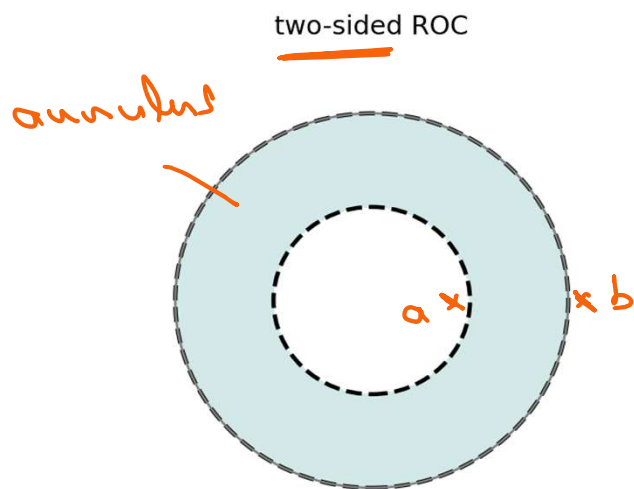
Nonzero only for $n \leq -1$.

**Same pole at $z=a$.
Different support.
Different ROC.**

ROC geometry: annuli bounded by poles

35-55 MIN

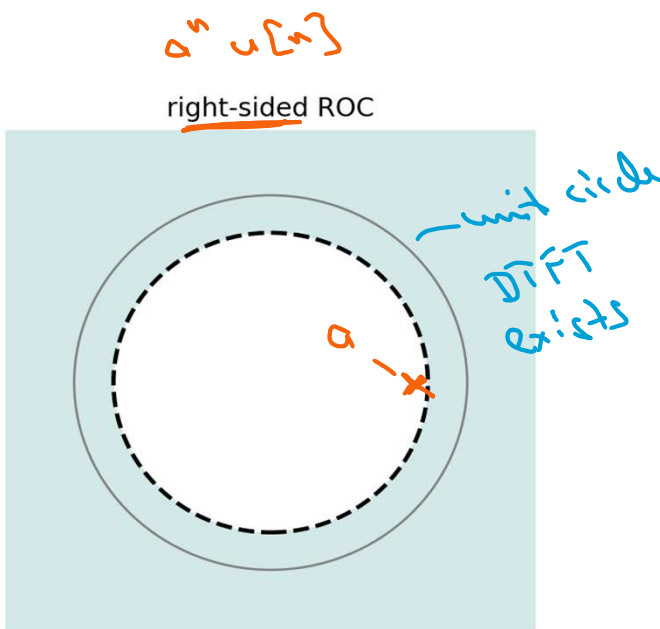
For rational transforms, the ROC cannot contain a pole.



Two-sided

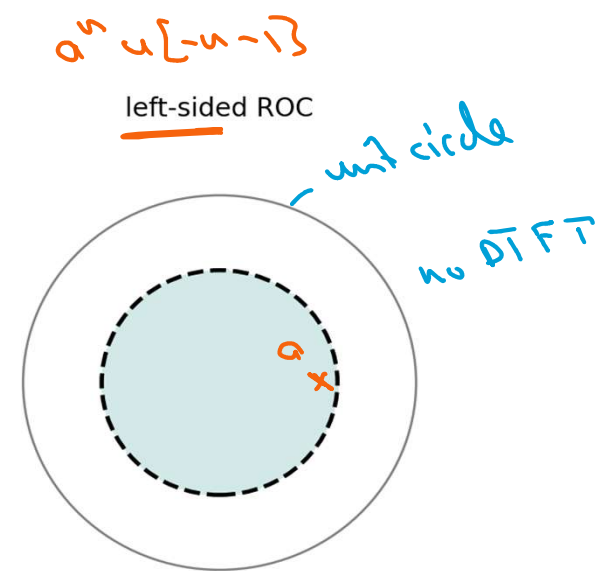
ROC lies between pole radii.

needs two poles



Right-sided

ROC is outside the outermost pole.



Left-sided

ROC is inside the innermost pole.

ROC rules worth being able to reason from

35-55 MIN

These are consequences of convergence, not arbitrary conventions.

1 ROC is a connected annulus centered at the origin.

2 ROC contains no poles.

3 Right-sided rational sequence $\rightarrow$ ROC outside outermost pole.

4 Left-sided rational sequence $\rightarrow$ ROC inside innermost pole.

5 Finite two-sided sequence: ROC is essentially entire z-plane except possible 0 or ∞ .

6 DTFT exists iff the unit circle lies in the ROC.

in general for two sided sequence ROC is annulus between an inner circle and an outer circle NOT containing any pole

Quick classification: infer support from the ROC

35-55 MIN

Think first; compute only if needed.

A $X(z)=1/(1-0.7z^{-1})$, $|z|>0.7$

pole $z=0.7$
right sided, DTFT exists

B $X(z)=1/(1-1.2z^{-1})$, $|z|>1.2$

pole $z=1.2$ right sided, no DTFT

C $X(z)=1/(1-1.2z^{-1})$, $|z|<1.2$

pole $z=1.2$ left sided DTFT exists

D poles at 0.5 and 0.9, $0.5<|z|<0.9$

two sided sequence no DTFT

Quick classification: infer support from the ROC

35–55 MIN

Think first; compute only if needed.

A $X(z)=1/(1-0.7z^{-1}), |z|>0.7$

Good answer: right-sided; DTFT exists

B $X(z)=1/(1-1.2z^{-1}), |z|>1.2$

Good answer: right-sided; DTFT does not exist

C $X(z)=1/(1-1.2z^{-1}), |z|<1.2$

Good answer: left-sided; DTFT exists because unit circle is inside ROC

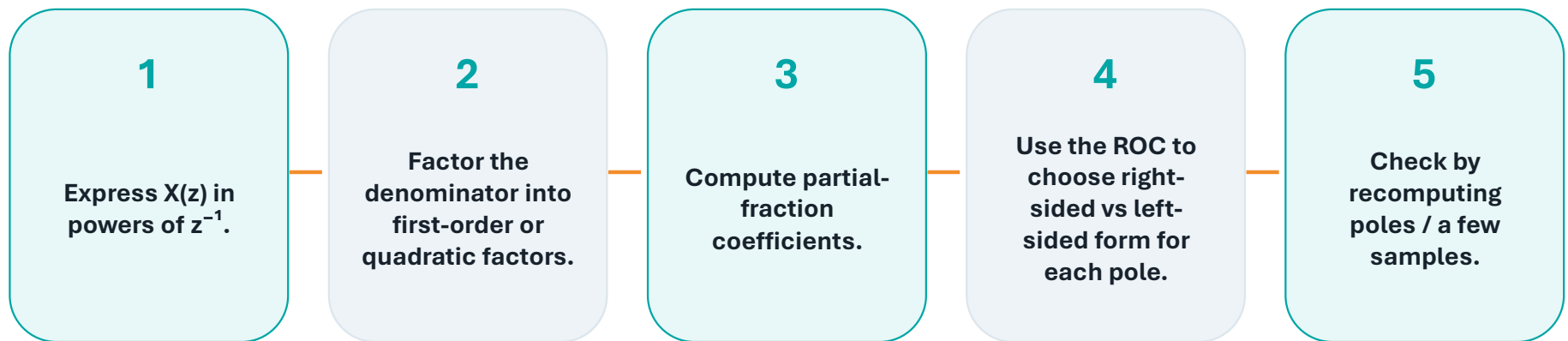
D poles at 0.5 and 0.9, $0.5<|z|<0.9$

Good answer: two-sided; DTFT does not exist

Inverse z-transform: practical workflow

55–75 MIN

For rational functions, partial fractions turns algebra back into transform pairs.

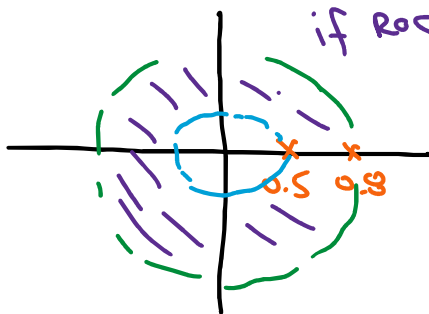


Do not invert the algebraic expression without the ROC: the same residues can map to different time-domain supports.

Worked partial fractions

55-75 MIN

One rational expression; multiple inverse transforms depending on ROC.



$$X(z) = 1 / [(1-0.5z^{-1})(1-0.8z^{-1})]$$

$$= -5/3 \cdot 1/(1-0.5z^{-1}) + 8/3 \cdot 1/(1-0.8z^{-1})$$

ROC $|z| > 0.8$

Both terms are right-sided.

$$x[n] = -(5/3)(0.5)^n u[n] + (8/3)(0.8)^n u[n]$$

ROC $0.5 < |z| < 0.8$

0.5-pole term is right-sided; 0.8-pole term is left-sided.

$$x[n] = -(5/3)(0.5)^n u[n] - (8/3)(0.8)^n u[-n-1]$$

residue of pole
at $z = 0.5$

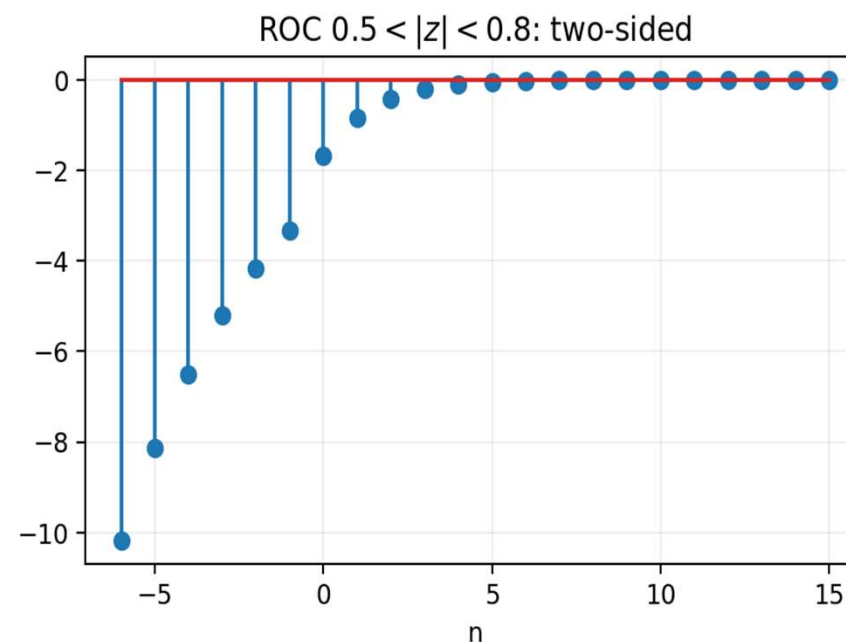
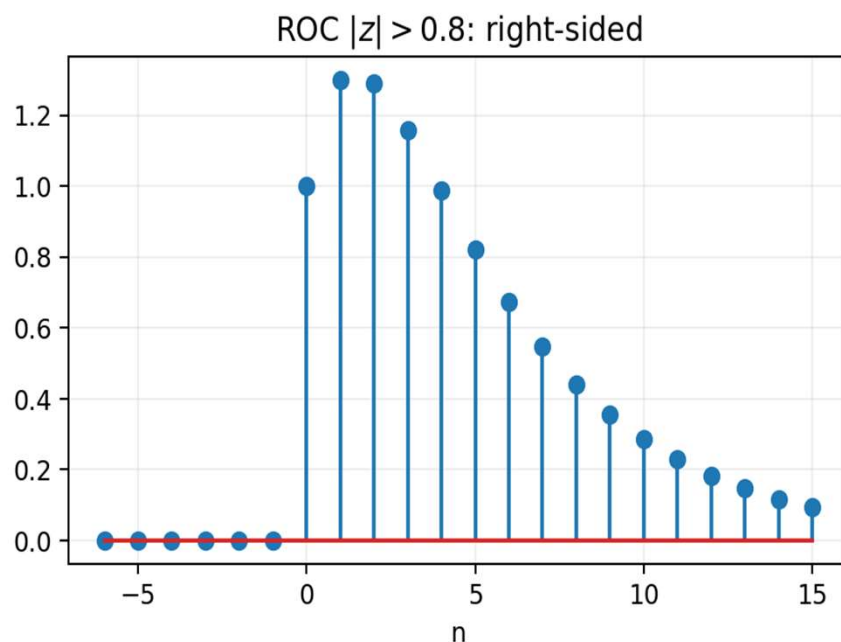
residue of pole
at $z = 0.8$

$$X(z) = \frac{1}{(1-0.5z^{-1})(1-0.8z^{-1})} = \frac{-5/3}{1-0.5z^{-1}} + \frac{8/3}{1-0.8z^{-1}}$$

The ROC changes the time-domain signal dramatically

55-75 MIN

Same poles and residues do not imply the same sequence.



This is why “ $X(z)=\text{rational function}$ ” is incomplete unless the ROC (or an equivalent support/causality condition) is specified.

Python demo: see convergence and residues numerically

55–75 MIN

Numerical tools confirm the analysis; they do not decide the ROC for you.

```
from scipy import signal

b = [1.0]
a = [1.0, -1.3, 0.4]
r, p, k = signal.residuez(b, a)
print(r, p)

# residues  $\approx [-5/3, 8/3]$ 
# poles = [0.5, 0.8]
```

What SciPy knows

- Algebraic residues and pole locations.
- A rational transfer-function representation.
- Frequency response if a stable interpretation is appropriate.

What SciPy does NOT infer

- Whether each term is right- or left-sided without an ROC/support condition.

Audio connection: poles forecast decay and resonance

55–75 MIN

Lecture 6 will turn ROC geometry into filter behavior.

Pole magnitude $|p|$

Sets exponential growth/decay rate in impulse-response terms.

Pole angle $\angle p$

Sets oscillatory frequency of a complex-conjugate pole pair.

Unit circle

Frequency response is evaluated here—if this circle lies in the ROC.

Preview: moving a pole closer to the unit circle makes the associated resonance sharper and the time-domain ringing last longer.

Concept check

CHECK

The expression and ROC must be interpreted together.

For $X(z)=1/(1-1.1z^{-1})$, which ROC gives a DTFT?

pole $z=1.1$

need $|z| < 1.1$
for ROC

left sided
to have DTFT

A. $|z| > 1.1$

B. $|z| < 1.1$

C. Both

D. Neither

Concept check

CHECK

The expression and ROC must be interpreted together.

For $X(z)=1/(1-1.1z^{-1})$, which ROC gives a DTFT?

A. $|z|>1.1$

B. $|z|<1.1$

C. Both

D. Neither

Correct: B. The unit circle $|z|=1$ lies inside the ROC $|z|<1.1$.

Common mistakes to prevent

These are high-value exam and debugging traps.

“The rational formula determines $x[n]$.”

false need ROC

“The z-transform always exists.”

No

“DTFT and z-transform are separate ideas.”

false $X(e^{j\omega}) = X(z) \big|_{z=e^{j\omega}}$

“A pole can lie in the ROC.”

NO!

Common mistakes to prevent

These are high-value exam and debugging traps.

“The rational formula determines $x[n]$.”

No—the ROC/support is part of the transform.

“The z-transform always exists.”

No—the ROC may be empty for some sequences.

“DTFT and z-transform are separate ideas.”

DTFT is the unit-circle evaluation when the ROC includes it.

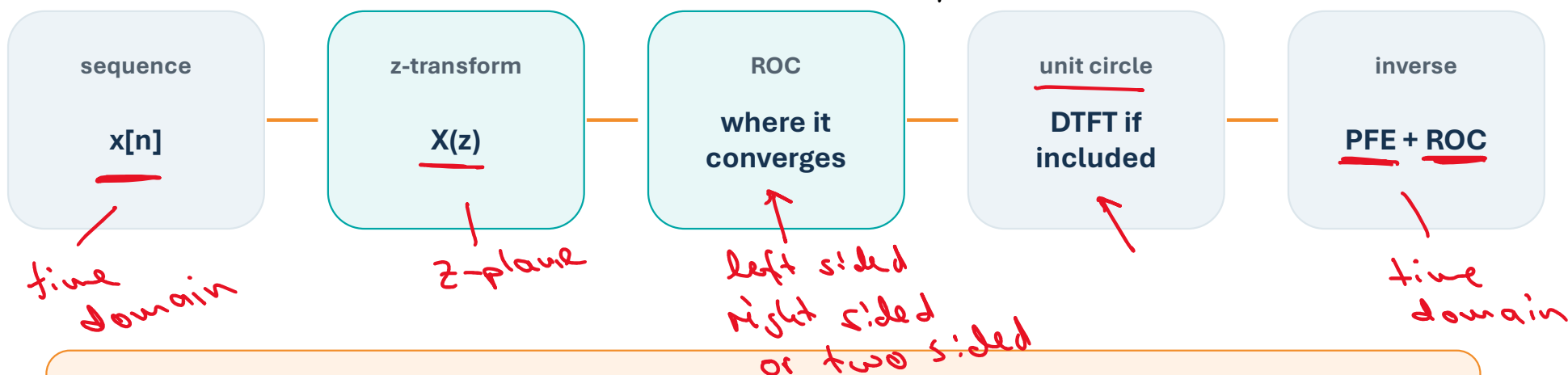
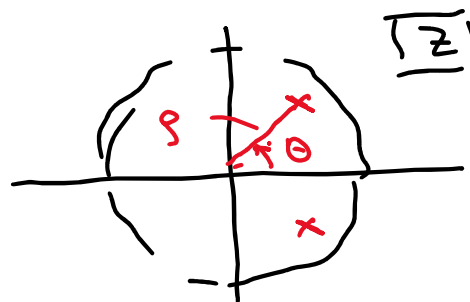
“A pole can lie in the ROC.”

Never: the transform diverges at a pole.

Lecture 5 synthesis

The z-transform is frequency analysis plus convergence geometry.

WRAP-UP



Exit ticket: Why can the same rational $X(z)$ correspond to different sequences?

complex conjugate pairs of poles
denominator

$$\rightarrow (1 - pe^{j\theta} z^{-1})(1 - pe^{-j\theta} z^{-1}) = 1 - 2p \cos(\theta) z^{-1} + p^2 z^{-2}$$

$$X(z) = \frac{1}{1 - 2p \cos(\theta) z^{-1} + p^2 z^{-2}}$$

Lecture 5 synthesis

WRAP-UP

The z-transform is frequency analysis plus convergence geometry.

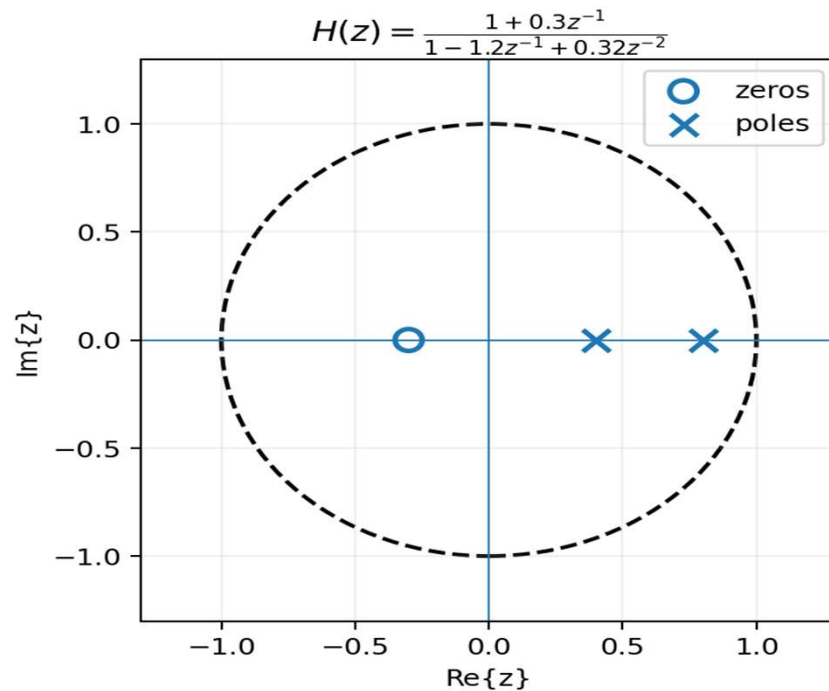


Exit ticket: Why can the same rational $X(z)$ correspond to different sequences? Good answer: because different power-series expansions converge in different ROCs, producing different time supports.

Next: transfer functions, poles, zeros, stability

NEXT

Lecture 6 turns the z-plane into a filter-design map.



- Difference equations become rational $H(z)$.
- Poles + ROC determine causality and stability.
- Zeros suppress nearby unit-circle frequencies.
- Poles enhance nearby frequencies.
- Audio filters become geometry you can see.